Electric Field Is Force On A Unit Positive Charge

E-Field PhET Simulation

> SYNOPSIS
>
> This note is about a "field." Specifically, the electric field. You should think about the field as a "sphere of influence." For example, if we are next to a planet like the Earth (like being on the surface as we are now), we are in its *gravitational* field, or its gravitational "influence." The strength of this influence is known as the gravitational field strength, g, which we otherwise know as the acceleration due to gravity, 9.8 m/s². So, what does a field do? **It accelerates stuff!** The gravitational field accelerates mass; in the same way, the electric field, which is a sphere of influence by a charge, accelerates another charge that happens to be close by. The electric field strength, or electric field, has the symbol **E**.

When we think about gravity, we focus on *a* mass and the force of gravity on the mass. For example, when we are at the Earth's surface, we "feel" the gravity as weight given by $F_G = mg$. We say the Earth pulls us. But we also know that we pull the Earth by the same magnitude towards us. We have the gravitational influence that the Earth feels, but we don't typically talk about this; we or other objects around the Earth that move because of the Earth's influence are the focus. **The idea of the field is to emphasize this influence** and facilitate figuring out the force on the object of interest due to the field source. If a charge q is in the sphere of influence of another charge, the force on q is $F_E = qE$; here, **E** is analogous to g, an influence generated by other nearby charges.

The following PhET simulation shows that every charge emits influences through electric fields represented by **field lines**. Another charge "takes a cue" from the lines and "figures out" the force on itself and, thus, how to move (accelerate). These lines have directions: **they originate from the positive charges and terminate on the negative charges.** You will place a +1 unit charge in the field generated by one or more charges. This +1 unit charge will conveniently reveal the direction and strength of the influence (i.e., force) from the charge(s) that generated the field. (This +1 unit charge is, thus, known as the **"test charge"** and, by default, *always* positive.)

PERFORMANCE EXPECTATIONS

1. **Draw** field lines emitted by positive and negative charges separately (single charges) or together (dipole).
2. **Describe** the strength and direction of the force a charge feels in the electric field using the test charge.

Instructions and questions

1. Type "electric field phet sim" into the Google search bar, select and run "Charges and Fields" PhET simulation.
2. Check the **Grid** box.

3. Pick a positive or negative charge from the box at the bottom of the screen and drag and drop it onto the grid. The sensor is your test charge that reveals the strength and direction of the force a +1 unit charge (the test charge) would experience.

Part I — point charge

4. Drag and drop a **+1 nC** charge onto the middle of the Positive Charge box below and **draw the field lines**. Note that relative field strength is indicated by the brightness of the arrows, not by their length, and your field lines should be continuous and indicate the directions.

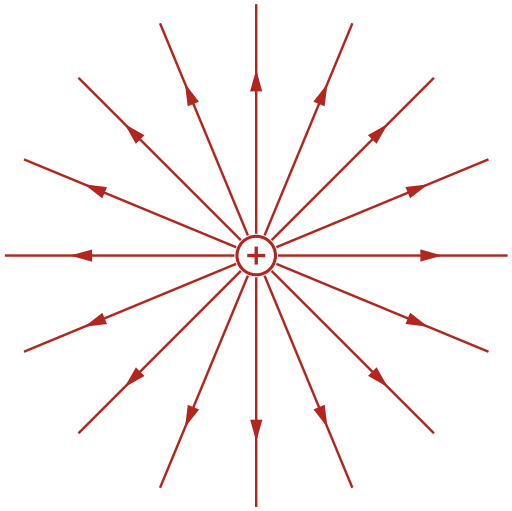


Figure 1. A single positive charge. The lines are straight, they are evenly spaced all round, and they all point away from the charge.

5. Clear the field and repeat the procedure with a **−1 nC** source charge.

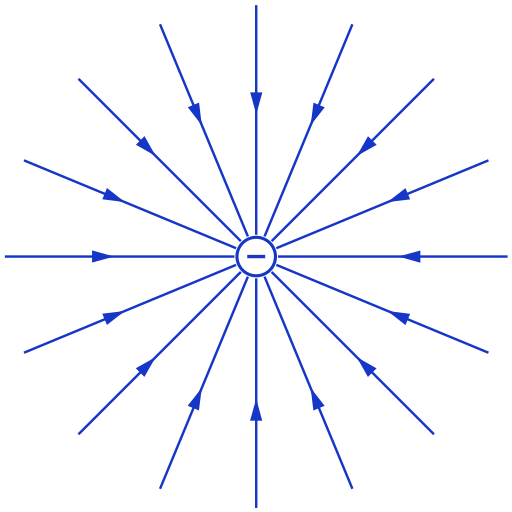


Figure 2. The same picture for a negative charge, with every arrow reversed. Same spacing, same straightness, opposite direction.

6. Drag and drop an **E-Field Sensor** and place it in the spots below at various distances from positive or negative source charge to measure the field strength (a “force”-like quantity represented by a **vector**). **Draw each location’s field strength vector** (this is the force a +1 unit charge feels).

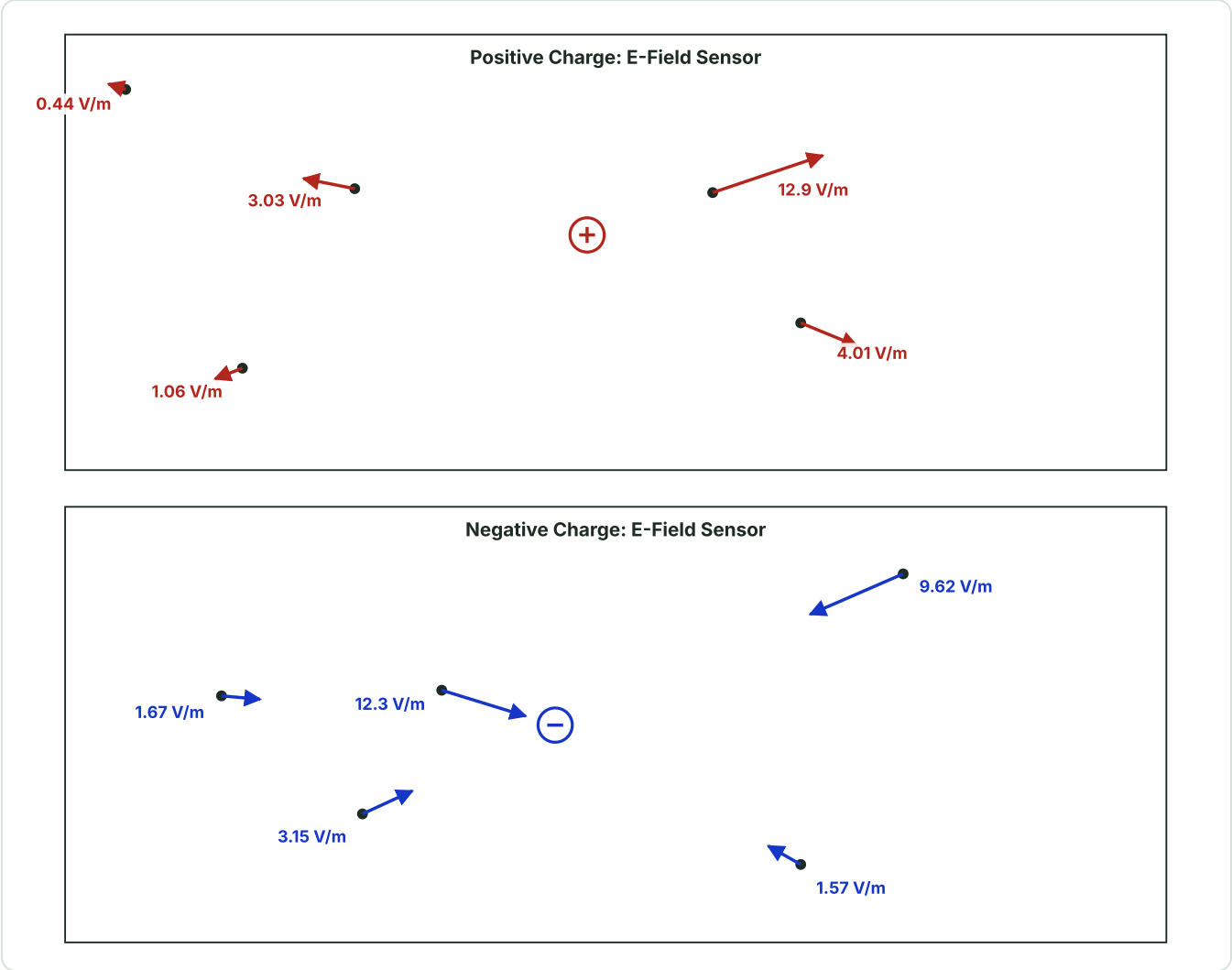


Figure 3. The ten readings taken in class, at the ten places they were taken. The sensor reads the force on the **test charge** — a +1 charge, always positive — so around the positive source every arrow points **away**, and around the negative source every arrow points **toward**. Look at the numbers, not just the picture: **12.9 V/m close in against 0.44 V/m out at the corner.** The far one is not a little weaker, it is nearly thirty times weaker. *Arrow lengths are compressed so the smallest stays visible.*

Part II — electric dipole (one positive and one negative charge close by in space)

7. What does the E-Field Sensor show?
   The **length** of the arrow: _____
   The **direction** of the arrow: _____

8. How do the E-Field Sensor arrows for the negative charge differ from those for the positive charge?

9. Drag and drop a **+1 nC** and **−1 nC** charge onto the field (the negative charge closer to the top of the field and the positive charge closer to the bottom). **Draw the resulting electric field** around the negative and positive charges. **Indicate the direction of the field lines! (Where do they begin, and where do they end?)**

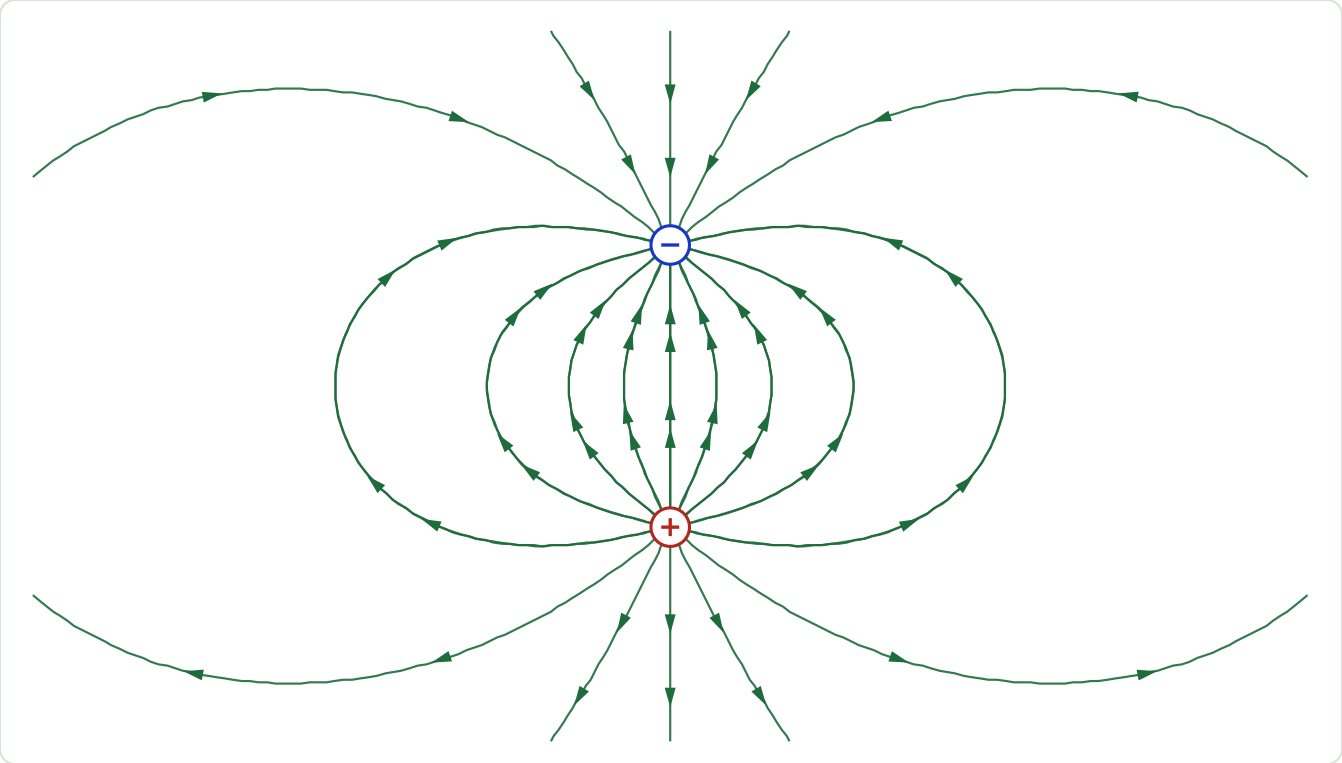


Figure 4. The dipole. Every line **begins** on the positive charge and **ends** on the negative one — there is nowhere else for them to start or stop. The lines near the middle run almost straight across; the ones that leave sideways have to swing a long way round.

10. Move the charges closer. **What changes occur, particularly concerning the field strength farther away from the source charges and, thus, the <u>resultant</u> field strength (the sum of the two fields) in general?**

11. Looking at the field strength at a location far enough away from these source charges, what can you say about the field when they appear to be atop one another?

12. Lay one source charge directly on top of the other. What can you say happens to the field in general?

Part III — two like charges close by

13. Reset the field and drop **two negative charges in a vertical line**, with one charge near the top and the other near the bottom. **Draw the resulting E-Field.**

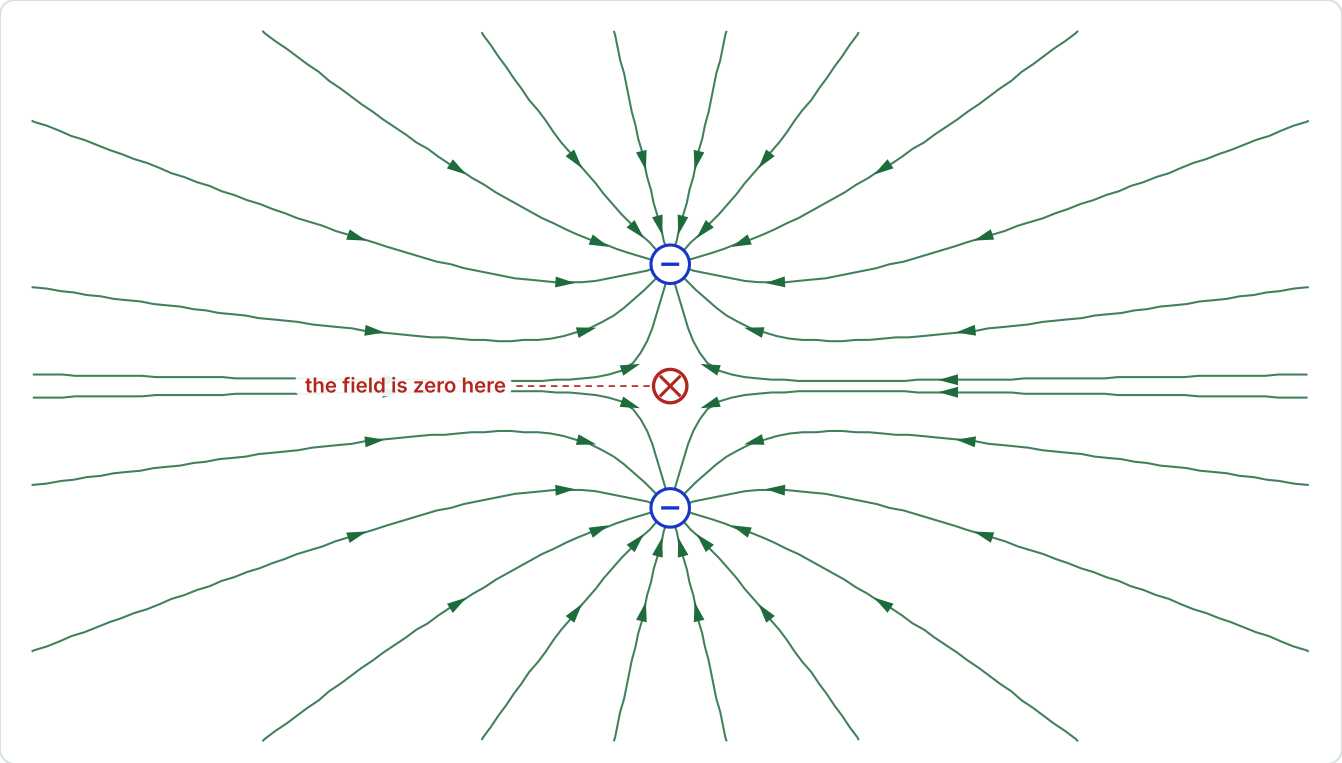


Figure 5. Two like charges. Now there is nowhere for the lines to begin, so they all come in from far away. Between the two charges the two fields pull in opposite directions — and **exactly midway** they are equal and opposite.

14. Is there a place where the field equals zero? Why is or isn't there such a spot?

The same three cases, as graphs

You have been drawing field *maps*. Here is the same information plotted along the line that runs straight through the two charges — the force a +1 test charge feels at each point, with rightward counted as positive. F_1 is the force from charge 1 alone and F_2 the force from charge 2 alone; what the test charge actually feels is the two of them added.

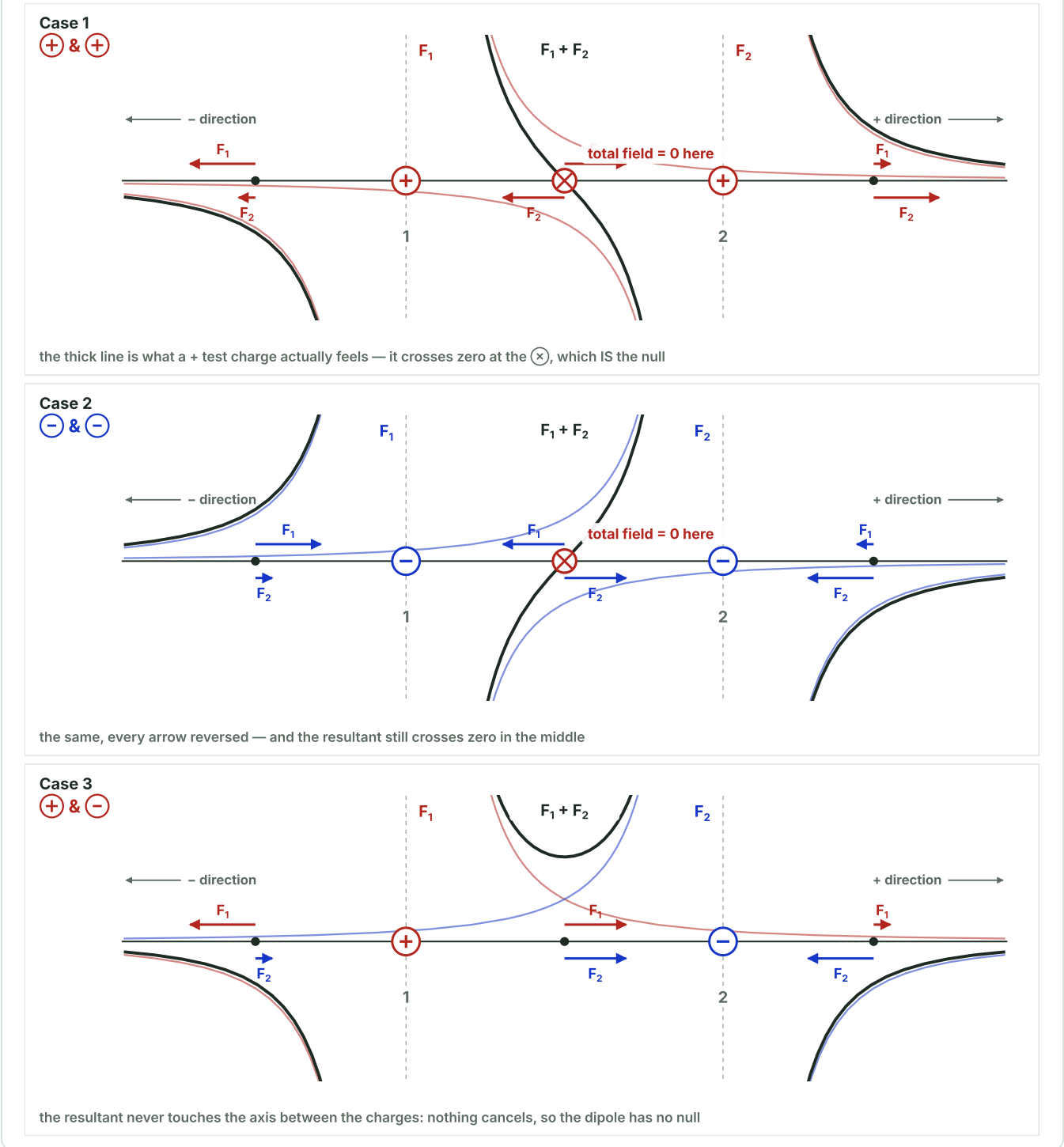


Figure 6. Up on these plots means the force pushes to the RIGHT — the convention is written on each axis. The black dots are three places to stand, and the two arrows at each one are what charge 1 and charge 2 do to a + test charge put there. **Cases 1 and 2 are the like-charge pairs:** outside, both arrows point the same way; between the charges they **oppose**, and at the midpoint they are equal — **that is the ⊗ you just found. Case 3 is the dipole:** between the charges both arrows point the **same** way, so nothing can

cancel, and the dipole has no null anywhere. *Arrow lengths are drawn to a compressed scale so the small ones stay visible — the nearer charge really wins by more than the picture shows.*

And read the thick black line first. The two thin curves are what each charge would do *on its own*; the thick one is $\mathbf{F}_1 + \mathbf{F}_2$, which is what a test charge actually feels. In Cases 1 and 2 it **crosses the axis**, and the crossing is marked ⊗ — **that crossing is the null**, and it is the only thing on the page that proves one exists. In Case 3 the thick line dips between the charges and **never touches the axis**. No crossing, no null.

THE PUNCHLINE

E-fields are vectors

Almost everything in this note follows from that one sentence. Two fields at the same point do not take turns and do not average — they **add head to tail**. That is why a + and a − on top of each other give nothing, and why two negatives give a dead spot exactly in the middle. And the other keeper: **flux = line density**, which is the idea Gauss's law is built on later in this module.

Summary

WRITE YOUR FIVE FIRST

Summarize **five things** you learned about the electric field from this simulation. Write yours before you open mine — the point is what *you* took away.

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Looking ahead

You now have a picture of the field and a rule for adding fields. **Note 4** asks the other half of the question: not what force a charge feels, but **how much energy it carries** — and because energy is a scalar, that turns out to be the easier half.